Investigating the Interpretability of BERT for Emotion Classification by Comparing SHAP, LIME, and Integrated Gradients

Final Report

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1. Introduction

1.1. Motivation

Modern machine learning models provide precise and accurate predictions across a wide variety of applications. As artificial intelligence continues to advance, these predictions become not only more accurate but also more complex. With increased complexity comes greater difficulty in explaining the internal decision-making processes of these models and interpreting final predictions. This can lead to distrust in a model's final output, which is influenced by factors such as bias and spurious correlations. In high-stakes applications like suicide prevention and mental health support, it is crucial to be able to interpret if a prediction was based on meaningful rather than irrelevant signals, so final decisions are not made based on an unexplainable model. Explainability techniques can be used to conclude from a model's output what it is basing its decision on and whether it can be trusted.

1.2. Problem Statement

Modern machine learning models that perform emotion- and sentiment-based prediction for text-centred online content often operate with a lack of transparency and credibility. This can result in distrust of a model's output due to inherent bias, setting a worrying precedent at a time when critical systems require these models to keep online communities safer (e.g. moderating social sites to prevent hate speech and bigoted language). In this project, our objective is to investigate and experiment with various interpretability methods for AI to determine which is best at explaining BERT's predictions. We will be using three methods: **LIME**, **SHAP**, and **Integrated Gradients (IG)**. These methods were chosen to analyze and compare perturbation- and gradient-based explanation methods. We aim to determine which method can generate explanations that make the reasoning for the BERT prediction clear and human-readable. Having a clear output explaining the reasoning helps to understand whether the model has predicted a particular label for the right reasons.

1.3. Research Questions

In this report we aim to learn how explainability methods compare in how they measure importance and impact by tokens on the prediction associated with a text sample by BERT, by answering the following research questions:

1. How do the explainability methods agree and differ in their measurement of importance based on token impact on BERT's emotion classification predictions?

2. Do the explainability methods agree on the top-n most impactful words for an emotion label, based on Jaccard similarity?

3. Are the most impactful tokens parsimonious; are the importance scores focused on a few clear words, or do the tokens all have similar scores?

4. How does the gradient-based Integrated Gradients method perform against the perturbation-based methods SHAP and LIME?

2. Background

2.1. BERT and Transformer Models

Bidirectional Encoder Representations from Transformers or **BERT** is capable of capturing contextual meaning when processing text for classification. BERT tokenizes words via the WordPiece tokenizer, capturing their meanings as vector embeddings [1] [2]. This project seeks to expand on a previous project by one of the authors where BERT was fine-tuned on the GoEmotions dataset for the purpose of emotion classification. This project uses Explainable AI methods to understand why the BERT model made predictions as it did.

2.2. GoEmotions Dataset

The GoEmotions dataset has 58,000 text samples taken from Reddit comments, annotated by humans with 28 emotion labels[3]. The fine-grained labelling adds nuance and meaning when fine-tuning the BERT model, it provides a challenge in trying to use explainability methods. The GoEmotions paper used Ekman emotion families [4] to collapse the 28 labels into 6 basic categories for evaluation after model training. This method is used in this project as well, to overcome the challenge of trying to compare and evaluate multilabel data across 28 labels.

2.3. Explainable AI Methods

2.3.1. Explainability with SHAP

SHapley Additive exPlanations or **SHAP** assigns each token or work a Shapley value, corresponding with the calculated impact that the token has on the prediction by BERT. It does this through a process of perturbation, changing text samples one token at a time to measure the impact on BERT's score.

Equation The equation for finding the Shapley value ϕ_i for token i is:

$$\phi_i = \sum_{S \subseteq \{1,\ldots,p\}\{i\}} \frac{|S|!(p-|S|-1)!}{p!}[val(S \cup \{i\}) - val(S)], \tag{1}$$

where N is the set of all words in the sentence, demarcated by [SEP], S is a subset of tokens minus i, $val(S)$ is the prediction the model makes based on subset S, the marginal contribution $[val(S \cup i) - val(S)]$ is the difference in prediction when i joins subset S, and $\frac{|S|!(p-|S|-1)!}{p!}$ is the probability this subset will form.

Axioms SHAP is considered to be a fair way of evaluating contributions to a common goal as it satisfies four axioms:

1. **Efficiency:** The prediction score made by the model is equal to the sum of Shapley values for each token.
2. **Null Player:** A Shapley value of zero is accorded to a token with no contribution.
3. **Symmetry:** If the model's prediction is the same when two tokens are swapped, they have the same Shapley value.
4. **Additivity:** Two summed evaluations by a model will sum their Shapley values for the same token.

PartitionExplainer `PartitionExplainer` is an algorithm used by the implementation of SHAP in the project to approximate the Shapley values. Instead of doing the computationally expensive task of evaluating each token individually, `PartitionExplainer` approximates the Shapley values with Owen values, which treat the tokens as groups that it iterates through recursively. This allows SHAP explanations to be computed in polynomial time as opposed to many SHAP algorithms, which take exponential time.

2.3.2. Explainability with LIME

Local Interpretable Model-Agnostic Explanations or **LIME** uses a surrogate model trained on perturbed inputs to observe changes in the output [5]. **LIME** can be used to produce an explanation for each feature's contribution to a model's final output and can be applied to a wide range of models due to its model-agnostic nature. LIME does this by exploring specific instances of a model's predictions and perturbing them, masking features and combinations of features in that instance to examine how the model's output changes. This information is used to fit an interpretable surrogate model that approximates the behaviour of the original model around that data instance. This provides an explainable output for feature contribution [6].

Equation To fit the interpretable model around an instance x, **LIME** minimizes a loss function represented as follows:

$$\xi(x) = \underset{g \in G}{\arg\min}\ \mathcal{L}(f, g, \pi_x) + \Omega(g), \tag{2}$$

Where f is the black box model, g is the interpretable surrogate model, G is the set of possible interpretable models, $\Omega(g)$ is a value given to g to regulate excess model complexity, and π_x the weight of how similar a perturbed instance of text is the to the original, which defines a local neighborhood. The weighted loss $\mathcal{L}(f, g, \pi_x)$ uses these variables to measure how close g approximates to f on the perturbed instance.

Importance of Locality and Perturbation Locality in the context of **LIME** refers to not only the exploration of a specific instance of data, but also the approximation of perturbed samples to create a **local neighborhood**. For text based data, perturbations are generated by masking subsets of tokens from the original instance and obtaining a prediction from the black-box model f without the effects of those features. Perturbed samples are not equally influential, as masking a single influential token can greatly sway a final prediction. This implies the perturbed instances most similar to the original are most important in training the surrogate model. By assigning a weight of proximity between samples, LIME defines a neighbourhood, ensuring it works with similar samples [7].

Kernel Width and Stability While a small, local neighbourhood is desirable, sampling such a narrow range can produce volatile results as high weights are assigned exclusively to proximal samples. A larger neighbourhood can produce more stable results, but risks explaining samples that differ too much from the original instance, skewing the locality of results. Kernel widths assign an importance to perturbed samples proportional to their distance. A small kernel width assigns greater importance to close, similar samples, while a large kernel width assigns roughly equal importance regardless of distance. A stable model is one whose token weights do not drastically vary when the kernel width is changed. To achieve an optimal neighbourhood, results should be as local as possible, implying the smallest possible kernel width that still produces stable token weights across a given range [8].

2.3.3. Explainability with Integrated Gradients

Integrated Gradients explains a model's prediction by attributing importance scores to each token. It accumulates gradients along a straight path from a user-defined baseline input (e.g. all zero vector or [PAD] token representation of words) to the actual input [9].

Equation Integrated Gradients is represented by the following path integral:

$$\text{IntegratedGrads}_i(x) := (x_i - x_i') \times \int_{\alpha=0}^{1} \frac{\partial F(x' + \alpha \times (x - x'))}{\partial x_i} d\alpha, \tag{2}$$

where $\text{IntegratedGrads}_i(x)$ is the integrated gradient for the i-th input feature, x is the actual input, x' is the baseline input, F is the function of the neural network, and α is a parameter that varies from 0 to 1 on a straight path.[10]

Axioms This path integral for Integrated Gradients satisfies the following fundamental axioms:

- **Completeness**: The importance score is the difference between the prediction for the sample and the prediction for the baseline input.
- **Sensitivity**: Any token that has an impact on the prediction of the model receives a non-zero importance score.

3. Methodology

3.1. Environment and Model Setup

The environment was configured to use a local GPU via CUDA for computationally intensive tasks. Finding the computational resources necessary to run BERT efficiently was initially a challenge. To address this, the fine-tuned model weights were downloaded and migrated to a local Jupyter Notebook from their initial Google Colab environment.

3.1.1. Architecture

The BERT model used is the same as the model from the previous project, the bert-base-uncased sourced from Hugging Face. The uncased version was selected for its insensitivity to casing. The GoEmotions data set contains many samples with inconsistent casing, as social media text often contains unpredictable capitalization. While casing can contain important information, such as emphasis, the erratic nature of casing in social media makes the uncased model better suited. BERT is available in both large and base configurations, containing approximately 340 million and 110, respectively. While the large model offers an improved ability to capture complex patterns, the computational efficiency is directly proportional to this complexity. To this end, the base model was selected as a compromise between precision and computational efficiency.

3.1.2. Hyperparameters

The following user-controller hyperparameters were selected for the fine-tuning of BERT:

- **Optimzer:** The **Adam with Decoupled Weight Decay (AdamW)** optimizer allows for more flexible tuning of hyperparameters by preventing overfitting of the model

- **Learning Rate:** A base learning rate of 1×10^{-5} was used during the fine tuning process. Using a scheduler, this was increased linearly at a rate of 0.1. For BERT, small learning rates are typical to reduce the risk of exploding gradients and improve convergence speed.
- **Batch Size:** During the fine-tuning process, a batch size of 16 was used. This value provides balance between computational efficiency, convergence speed and model stability.
- **Epochs:** The model was fine-tuned for 5 epochs.
- **Loss Function:** During training, a weighted binary cross-entropy loss was used via the `BCEWithLogitsLoss` function. This function improves training stability by combining a sigmoid layer with Binary Cross-Entropy. To improve classification of infrequent emotion labels, the `pos_weight` hyperparameter, defined as the ratio of negative to positive samples for each label was passed to this function.

3.2. Data Pipeline

Since LIME, SHAP and IG work to explain the output of BERT, they use a shared data processing procedure to transform the GoEmotions dataset into the necessary format. This includes tokenization using WordPiece and a conversion of the dataset from CSV to a tensor representation.

3.2.1. Tokenization with WordPiece

Using the WordPiece algorithm, the raw text of each sample in the dataset is broken down into words and subwords, resulting in tokens. These tokens are assigned a vector ID based on its index in a fixed vocabulary used by BERT. Special tokens are used to signal the structure of the text to the model. The token [CLS] is used to represent the beginning of a text sample, [SEP] to mark the separation of sentences, and [PAD], a null token that fills samples to ensure all samples are all the same length.

3.2.2. Pytorch

WordPiece tokens need to be mapped to three kinds of tensors:

- **Input IDs**, which represent the token indices from the fixed vocabulary
- **Attention masks**, which will allow the model to ignore [PAD] tokens using a binary value, and
- **Token type IDs**, which will mark which sentence in a sample the token belongs to.

These tensors are encapsulated within the **DataLoader** class, which handles iteration and batching. An initial batch size of 16 was used for testing and later increased to 64 to better utilize the GPU. Since the model is initialized in evaluation mode, increasing the batch size does not affect performance metrics such as the F1-score, but significantly improves the computational efficiency of the XAI explainability methods.

3.2.3. Ekman label mapping

The process of using explanation methods on a multilabel dataset with 28 labels posed a challenge for multiple reasons: there are too many labels to look at individual comparisons of each one, and the labels are not all represented equally so sampled data might miss out entirely on rare emotions

such as "grief". To overcome this problem we used the 6 main Ekman emotion groups [4] to map the 27 emotions excluding samples that only had a "neutral" label. So labels like "grief" and "sadness" are grouped in one label, improving on the issue of having a scarcity of samples in a label group. After collapsing the 27 labels into 6 they can be sampled, explained, and compared more effectively.

3.2.4. Iterative Stratification for Sampling

In a dataset with 58k samples, it would not be computationally feasible to run the explanations on every BERT prediction for those samples. The **iterative stratification** method was chosen for its ability to take representative samples of multilabel data. It does this while preserving proportions of label combinations, not just labels themselves [11]. This ensures that the sample we take to explain predictions made by BERT will exhibit the same patterns and correlations as the overall dataset.

3.3. Methodology of SHAP

3.3.1. Implementation of SHAP

Once the data has been sampled, it is translated from input IDs back to raw text, which SHAP needs to explain BERT's predictions.

Wrapper Function The wrapper function converts perturbed samples from raw text to input IDs and attention masks that BERT can process in a forward pass. Logits representing the probabilities are returned, ready for evaluation.

PartitionExplainer The `PartitionExplainer` algorithm is called to give the importance scores of each token in each sample taken with iterative stratification. Treating groups of tokens as if they were individuals and iterating through the ones that are estimated to be most important allows the `PartitionExplainer` algorithm to improve time complexity. The output importance scores are saved to a CSV for direct comparison with other explainability methods.

3.3.2. Visualization

To directly compare the three methods, the importance values are displayed in a 3D scatterplot, which shows the importance score generated by each method on each axis. Barplots of the top tokens by importance score are also generated, as well as the Jaccard similarity scores of the top-n words for each method, displayed in a connected scatterplot. As well as the comparison visualizations generated, Shapley values for some text samples are visualized using the `shap` library in Python. In the text plots generated in testing, the Shapley values of each token determine the colour and intensity of highlighting over tokens in their raw text form.

3.4. Methodology of LIME

3.4.1. Implementation of LIME

To interpret across all possible emotion labels from the dataset, a LimeTextExplainer is initialized with the list of the 6 Ekman emotion groupings.

Wrapper Function As another model-agnostic method, LIME utilizes the same wrapper function used by SHAP to interact with BERT's output behavior. After accepting raw text as an input, the function tokenizes then converts it to a tensor of token IDs and attention masks. These outputs are applied to a sigmoid function to convert logits into probability scores. LIME can explain a single instance using this wrapper function, resulting in a 6-dimensional vector where each dimension corresponds with a probability score for a given Ekman emotion group.

Explaining an Instance The wrapper function is passed to LIME to generate an explanation for a sample of text. The resulting vector of probability scores represents the likelihood of BERT predicting an emotion label. The index corresponding to an Ekman emotion group is selected from the probability vector and passed to the explainer. The explainer computes feature contribution for that label by comparing it to perturbed samples in the local neighbourhood. Since LIME explains individual instances, this process is repeated for all 1000 samples taken using iterative stratification.

3.4.2. Instance Visualization

LIME explanations can be visualized using a built-in rendering. The feature contributions are represented in two ways:

- A highlight over each word in the instance, where stronger highlights indicate a larger magnitude of contribution towards the selected emotion label.
- A bar plot, where contribution scores are displayed and ranked by magnitude.

These visualizations allow for qualitative analysis of whether BERT is making predictions for appropriate reasons.

3.4.3. Comparison Visualization

Visualizations of importance values for the top-n tokens are generated and displayed using barplots. Jaccard similarity scores are generated to analyze similarities of these results between each explainability method. Jaccard comparison plots are generated using these scores, along with a 3D scatterplot to compare individual token importance for each method. These visualization methods allow for comparison and analysis of how each method differs in it's approach to explainability.

3.5. Methodology of Integrated Gradients

3.5.1. Implementation of Integrated Gradients

Forward Function The forward function processes input data through the neural network. Logits that represent prediction values are output from the forward function. Input IDs and attention masks are taken as input [12].

Baseline Input Next is the generation of the baseline input. The baseline input, is a sequence of [CLS], [SEP], and [PAD] tokens that together represent an empty input of the length of the text sample. The sequence begins with [CLS], is padded with [PAD], and [SEP] tokens that are placed corresponding to sentence breaks in the input IDs. A custom function is used to create the baseline based on the input IDs in the sample of GoEmotions data. The baseline is then evaluated for feature attribution.

The Evaluation Loop The evaluation loop will analyze the 1000 textual examples in the GoEmotions dataset, provided by the iterative stratification step. This step is repeated once per emotion label, demonstrating the necessity of the Ekman mapping step. The evaluation loop calculates the importance score for each token in each sample. This step prepares these scores for each sample as data to be saved and visualized, in comparison with the other two methods.

3.5.2. Visualization

As well as the comparison visualizations that are generated to test the importance scores and similarity with the other methods, visualizations are used to test the validity of the data and pipeline. Within the evaluation loop, visualizations are generated to explain each piece of text. The visualization first produces a textual output that outlines the current example and the top predicted emotion found within the input, along with the confidence score from BERT's prediction. The true label of each emotion, the importance scores, and the display of the raw text with colour highlighting corresponding to the intensity and sign of the importance score are displayed for each sample.

3.6. Evaluation Metrics

In our comparison analysis, we use Jaccard similarity and a comparison of importance scores to assess each explanation method at quantifying token importance.

3.6.1. Importance Score Comparison

In both the Top Tokens by Importance Score and Importance Score Distribution Sections, importance scores of individual tokens are used to generate comparative visualizations. Each explanation method outputs an **importance score** for each token between -1 and 1. The tokens with the highest magnitude of impact on the output prediction by BERT have the largest absolute value importance score. A positively-signed importance score pushes the BERT prediction toward a target emotion label, making it likely to have a higher degree of confidence, while a negatively-signed importance score pushes the prediction away from the label.

In Section 4.1 Top Tokens by Importance Score, the top 20 tokens are displayed alongside their importance scores in barchart format for each of the three explanation methods. In Sections 4.3 Importance Score Distribution, the importance score obtained by each of the three methods is plotted at once for each sample by label, using 3 axes to represent the three explanation methods in a 3D scatter plot [13].

Different areas of the output graph will correspond to varying degrees of agreement or disagreement by the methods on the importance magnitude of various tokens. The line where $x = y = z$ describes perfect agreement between the three methods. Outliers from this line show where the methods disagree on those tokens. The closer a point is to the origin, the smaller its importance score, while the tokens farther from the origin have a greater importance score.

3.6.2. Jaccard Similarity

When measuring how similar the top-n tokens are between different explanation methods, **Jaccard similarity index** is used as the measurement of agreement between the resulting sets. When looking at two sets of top-n tokens, Jaccard similarity is the ratio between the intersection and the union of the two sets, or $J(A, B) = \frac{|A \cap B|}{|A \cup B|}$. This results in a number between 0 and 1, with 0 being no agreement at all and 1 being perfect agreement between the two sets [14]. This project evaluated

the Jaccard similarity index for the top-n tokens of each explanation method with n between 5 and 50, capturing the most impactful tokens by importance score. For $n > 50$, the importance scores, at least for LIME and SHAP, drop off sharply to < 0.2, signifying less impact by those tokens on the predictions by BERT.

4. Results

4.1. Top Tokens by Importance Score

The top-n tokens, sorted by their importance scores, are collected to perform an initial qualitative assessment for their face validity. Upon examining the tokens, face validity appears high as many keywords seem to pertain to the target label. It can be seen that the importance scores for SHAP and LIME drop sharply for $n > 20$ while the IG importance scores decrease linearly and are higher for greater n, as seen in Figure 1 on the following page. For the complete set of plots displaying the top 20 words by each of the 6 Ekman labels, see Figures 4 to 8 on pages 20–22.

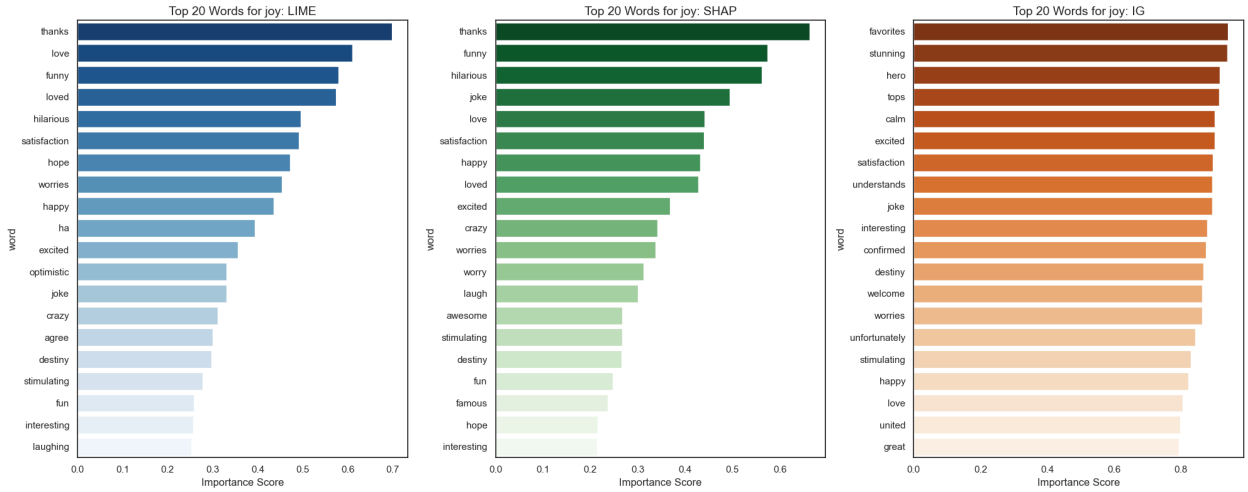

Figure 1: Top 20 tokens for the target label "joy" sorted by importance score evaluated by: *Left:* LIME. *Center:* SHAP. *Right:* Integrated Gradients.

Comparing the importance scores of the top-n words by their Ekman group shows that certain emotion labels like "surprise" in Figure 8 on page 22 seem to have fewer words that can move the BERT model's confidence in a prediction. The smaller number of tokens with an importance score > 0.2 is evidence of a label having fewer clear semantic indicators, or it is the explanation methods inability to pick up on the cues important to the model.

4.2. Jaccard Similarity Comparison

For the comparison of Jaccard similarity scores, they were generated for the top-n words with the greatest importance score about a target label, with $5 \leq n \leq 50$. Scores were generated to compare the sets created by each explanation method. The scores for each method comparison were plotted as shown in Figure 2, with the blue line corresponding to the Jaccard scores for LIME compared with SHAP, the orange line corresponding to the scores for LIME compared with Integrated Gradients, and the green line corresponding to the scores for SHAP compared with Integrated Gradients.

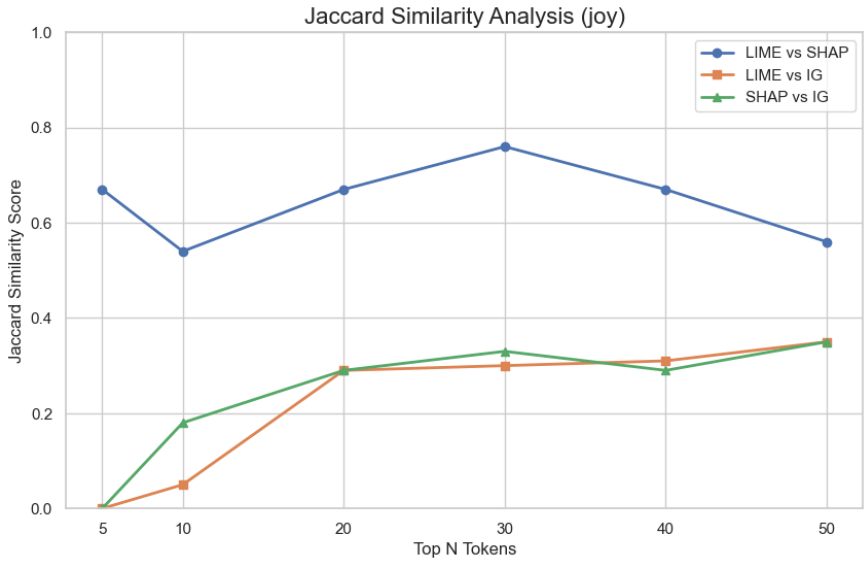

Figure 2: Jaccard Similarity Index for top-n tokens by importance score between LIME and SHAP in blue, LIME and IG in orange, and SHAP and IG in green, for the label "joy".

The comparison as seen in Figure 2 shows that LIME and SHAP have a higher degree of agreement or similarity between their top-n word sets, as the Jaccard similarity score is always greater than 0.5. The comparisons between LIME and Integrated Gradients, as well as SHAP and Integrated Gradients, show some agreement but never score more than 0.4 Jaccard similarity. Across the 6 Ekman emotion labels, this pattern of similarity between SHAP and LIME is consistent. The similarity between SHAP and Integrated Gradients is always less pronounced, as is the similarity between LIME and Integrated Gradients.

Notably, "surprise" is an emotion label that showed little agreement between the methods used. SHAP and LIME still have consistently higher Jaccard similarity scores than than the comparisons with Integrated Gradients, but all similarity scores are below 0.4, as shown in Figure 13 on page 25.

4.3. Importance Score Distribution

3D scatter plots are used to visualize a quantitative comparison of the three explanation methods. They do so by using the importance score calculated by each method for each of the three axes.

The cloud created by the scatter plot for each label is similar, forming a dense cloud at and near the origin, where most of the tokens, the neutral stopwords, fall. Each of the 6 Ekman emotion labels has a group of words in a sparse cloud along the line where $x = y = z$. While the LIME and SHAP axes show mostly positive importance scores, Integrated Gradients has a cloud of points along its negative axis as well. This results in a long and narrow cloud along the Integrated Gradients axis of the 3D scatter plot, with a "wing" along the positive part of the line $x = y = z$, as shown in Figure 3 and in Figures 14 to 17 on pages 26–29.

The compared importance scores for the label "surprise", as shown in Figure 18 on page 30, provided an exception to the pattern seen in the other Figures. This label had few words outside of the cloud around the origin, and they don't form the characteristic "wing" along the line where $x = y = z$.

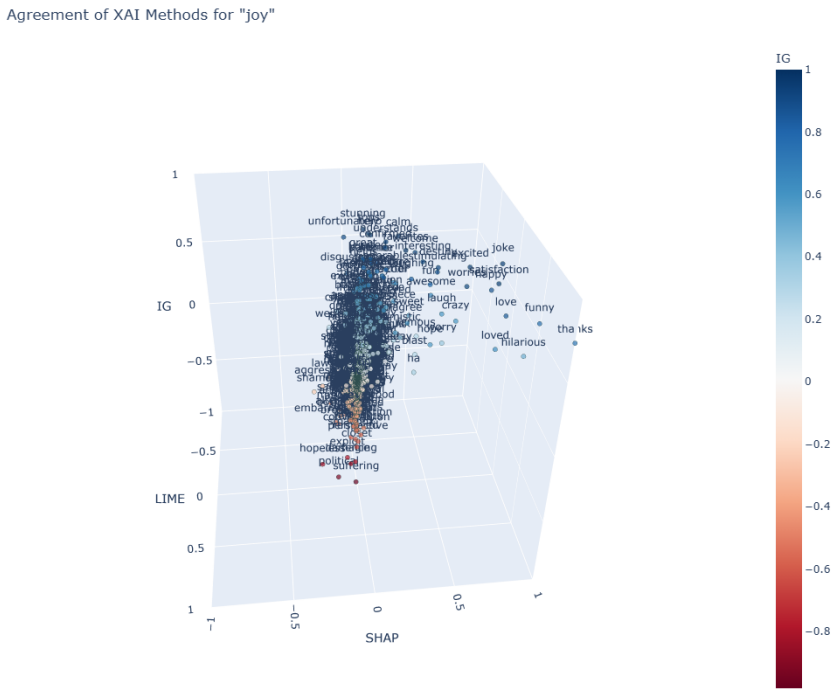

Figure 3: 3D scatter plot for the label "joy" with IG importance scores as the vertical axis, LIME importance scores as the axis along the depth of the plot, and SHAP importance scores as the axis spanning the width of the plot. The colour gradient follows the IG importance scores.

5. Conclusions

In this project we quantitatively compared three explainability methods, SHAP, LIME, and Integrated Gradients, applied to a BERT model fine-tuned on the GoEmotions dataset for emotion classification. By evaluating top-n words, Jaccard similarity, and importance score distributions, we can investigate the three methods for their parsimony and accordance in opinion.

5.1. Findings

Addressing Research Question 1: For each Ekman emotion label, there is strong agreement, particularly between LIME and SHAP, on both tokens demonstrating positively signed importance and those that are neutral. There is less agreement on the tokens that are negatively signed, which IG calculates many of, while there are few for SHAP and LIME. Importance is distributed among tokens differently among the three methods. The perturbation methods concentrate importance on a small subset of the top-n tokens while IG distributes importance broadly.

The neutrality of tokens to LIME and SHAP is evidenced by the cloud in the 3D plots that occupies the origin and how it is stretched along the IG axis. These points are evaluated with little to no importance score by SHAP and LIME, when IG calculates a range of scores. Negatively signed importance scores are more common for IG as well, which has many of varying magnitudes while SHAP and LIME have few and they are of smaller magnitudes. Negatively signed scores correspond with tokens that tend to push BERT away from predicting a label in a sample. This shows there is a difference in the calculated importance score between the gradient- and perturbation-based

methods.

On tokens that are evaluated by both SHAP and LIME as having greater importance, IG agrees as there thin clouds of values along the $x = y = z$ line for each Ekman label, as in Figure 3 on the previous page and Figures 14 to 17 on pages 26–29. The 3D scatter plot for "surprise" as shown in Figure 18 on page 30 presents an anomaly, as there is little agreement between any of the explanation methods, showing few clear word tokens identified by any method as important as well.

Addressing Research Question 2: Jaccard similarity scores of 0.5 or greater indicate strong agreement between SHAP and LIME, suggesting overlapping sets of top-n important words. IG tends to have significantly lower Jaccard scores of 0.4 or lower when compared with SHAP and LIME across all Ekman emotion groups, as seen in Figure 2 on page 12 and Figures 9 to 12 on pages 23–24. This indicates that IG identifies a much different set of important tokens, which is an observable pattern in the corresponding bar plots. The three methods tend to have increasing agreement as n grows large, suggesting that while IG differs from SHAP and LIME in identifying the most important tokens, there tends to be more overlap when considering a broader set of words.

"Surprise" as shown in Figure 13 on page 25 showed much lower Jaccard similarity scores than for the other emotion labels. All combinations were lower, including SHAP and LIME, which, while they were still greater than their respective comparisons with IG, remained < 0.4.

Addressing Research Question 3: Parsimony is reflected heavily in the results of SHAP and LIME, with importance scores dropping significantly after the top 20 tokens, suggesting that a small subset of tokens carries a disproportionate influence on the model's final predictions. This can be seen both by examining the top-n tokens directly in Figure 1 on page 12 and in Figures 4 to 7 on pages 20–21 and by examining the 3D scatter plots as the "wing" shaped cloud outside the large cloud around the origin $(0, 0, 0)$ encompasses the tokens that the methods agree are the top important words for that label.

In contrast, IG exhibits a linear decrease in importance scores as n increases, indicating that IG is more sensitive to the surrounding context than the perturbation-based methods.

"Surprise" is once again an outlier, showing two words for each SHAP and LIME that have an importance score > 0.2, as seen in Figure 8 on page 22. While the importance scores for IG are still higher and decrease slower than LIME or SHAP, they decrease rapidly compared to the other Ekman emotion labels.

Addressing Research Question 4: IG differed notably from SHAP and LIME in both the magnitudes of top-n tokens' importance scores, as well as the overall distribution of importance. While SHAP and LIME produced relatively sparse distributions of importance, IG assigned its importance scores broadly across the top tokens. IG frequently assigns negative importance values to tokens that SHAP and LIME often assign neutral values, suggesting that SHAP and LIME primarily identify the words that contribute positively to a prediction, while IG identifies tokens that detract from the prediction. IG's ability to identify contextual influences and nuance differentiates it from the perturbation-based methods, which instead tend to identify the most influential tokens in a text sample.

5.2. Conclusions

In conclusion, consensus matters for a task such as explaining predictions made by AI about emotions in text. This is because there is no fundamental "ground truth" when it comes to the

importance of a token on a model's prediction. It was important to explain the BERT predictions with multiple methods that had fundamental differences in how they calculated feature attributions. Nuance and a greater understanding of the functioning of the BERT model, and whether it functions at all for all emotion labels.

"Surprise" presented a challenge to explain for all three methods, even with sampling by iterative stratification and Ekman grouping of the emotion labels. This step may not have been sufficient to fix the underrepresentation of the label, as there are fewer than 5000 samples of "surprise" in the GoEmotions dataset. Also, this may be an emotion that has few words explicitly used to refer to it, especially when expressing it, compared to other emotions. "Surprise" may be a difficult emotion for BERT to predict, or it may not be specific tokens that are aiding in its prediction.

The perturbation-based methods SHAP and LIME perform differently at explanation than the gradient-based Integrated Gradients. The perturbation-based methods both seem to pick a few words on which to focus the importance scores. These two methods display high parsimony; the perturbation methods identify the few key words most influential on BERT's predictions. The Integrated Gradients method spreads importance widely, resulting in a much larger pool of words with high importance scores. This model is not parsimonious, but instead grants insight into the nuance and potentially the contextual information that the perturbation methods fail to capture.

All three AI explanation methods are useful for explaining the emotion classification predictions made by BERT. They have different aptitudes by themselves, and recommending one over the other would be based on context. For a context requiring concise human readability, SHAP or LIME would be appropriate, as they are parsimonious. For a context requiring understanding of the architecture and function of the AI model itself, Integrated Gradients is more appropriate as it gives a more nuanced look at the tokens and context that the model finds meaningful. All three have uses either because of their human readability or because of their detailed and nuanced view of linguistic meaning, and together their consensus is important in establishing trust and confidence in the predictions of an AI model.

6. Discussion

6.1. Inter-Method Consistency

SHAP and LIME have a greater Jaccard similarity score with each other than either do with Integrated Gradients, consistently for all the Ekman emotion groups. In part, this is because they differ in their ability to capture the tokens with a negative impact on the BERT prediction. Integrated Gradients evaluates clear groups of words that detract from a given label, while SHAP and LIME have very few negative attributions that are very small in magnitude. SHAP and LIME rely on perturbation to function, a model-agnostic method that does not access the internal gradients of the model. Perturbation methods see the surface-level functioning of the model, but may not be as sensitive as Integrated Gradients is to the impact of each token. Perturbation methods may also be unable to capture the ability of a token to detract from a label consistently, as Integrated Gradients does.

There are many words that Integrated Gradients identifies as important, which are neutral according to SHAP and LIME. This also detracts from the Jaccard similarity score when comparing LIME and SHAP with IG. Integrated Gradients spreads attribution over many tokens and is not parsimonious, while SHAP and LIME focus importance scores on the top-20 tokens. The gradient-based method seems more sensitive to words that do not have a large impact, but still have a measurable impact on the prediction made about a sample.

6.2. The Emotion Label "Surprise"

The emotion label "surprise" provided as an outlier for explanation method comparisons. Both the Jaccard similarity index shown in Figure 13 on page 25 and the 3D importance score comparison in Figure 18 on page 30 showed anomalous behaviour for this emotion label. The top-20 word analysis as shown in Figure 8 on page 22 displays only 2 words each that LIME and SHAP measure as having importance scores > 0.2; "curious" and "stunning" for SHAP, "curious" and "oh" for LIME. These are not explicitly about surprise, while many of the other labels contain themselves in the top-n words. Integrated Gradients can capture importance scores for the top 20 words of the label "surprise" but the actual word "surprise" does not appear in the list. This demonstrates that the way the emotion surprise appears in the English language, where it may depend more on contextual cues than clear linguistic ones, i.e. "I did not expect..." indicates surprise implicitly, whereas "I am angry that..." is a very clear and explicit way of indicating anger. It seems that the perturbation methods struggle with these implicitly expressed emotions. Integrated Gradients and other gradient-based methods that are more sensitive to the words that deter a prediction for a particular label, may be more appropriate for uses where this information is needed.

That "surprise" is hard to explain for the three methods may be an issue of label representation in the training data, as there were fewer than 5000 samples labelled "surprise" in the GoEmotions dataset. Despite the Ekman grouping used to navigate class imbalance, it does not appear to have solved the issue for this particular emotion label.

In the GoEmotions paper, the researchers mention that there was more disagreement between annotators regarding what should be labelled as "surprise" compared to other emotion labels [3]. This shows that "surprise" is an inherently difficult emotion to define, categorize, and identify by clear keywords, even for humans. The sampling issue is also not unique, as research from 2025 [15] mentions that rare labels such as surprise were more difficult for models like BERT to predict due to bias towards more common emotion labels.

"Surprise" may be inherently difficult for the explanation methods because it is not an emotion experienced as solely "good" or "bad", but depending on the context, can be either [16]. The perturbation-based explanation methods may focus on finding the words that have a positive association with "surprise", while the Integrated Gradients method may be more adept at capturing words that have a negative association with "surprise", due to its sensitivity to negatively signed importance scores. This conflict likely partially contributes to the low Jaccard similarity scores between all methods for the label "surprise".

6.3. Limitations

In the iterative stratification step, 1000 samples are taken to represent a dataset of 58000 samples. A greater sample size may improve the stability of results and show trends too subtle to show up in the experimental sample size. The sample size was limited by computational restraints, as the code to evaluate the importance scores was run locally and took several hours, depending on the method. The evaluation code was completed 6 times, once for each Ekman label, and then this whole process was completed for each method for a total of 18 computationally heavy code executions. The sample size and Ekman mapping made this part possible, but the computational restraints kept us from increasing the sample size.

Ekman emotion group mapping made the visualization of multilabel data possible, when the original dataset of 28 labels would be difficult to visualize effectively. Also, class imbalance results in too few samples of certain labels. BERT predictions may be affected by this bias, as would be the explanation methods. Grouping the 28 labels into the 6 main Ekman emotion groups helps

to adjust the data, as the rarer emotions are combined to avoid having too few samples. Also, 6 visualizations are human-readable, 28 plots would be too sparse and too many to parse to make sense of trends. The benefits of using Ekman emotion mapping are taken into account; it is an older method of grouping emotions, and in future work, it may be useful to examine other grouping methods.

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A. Supplementary Visualizations

A.1. Top 20 Important Words by Label

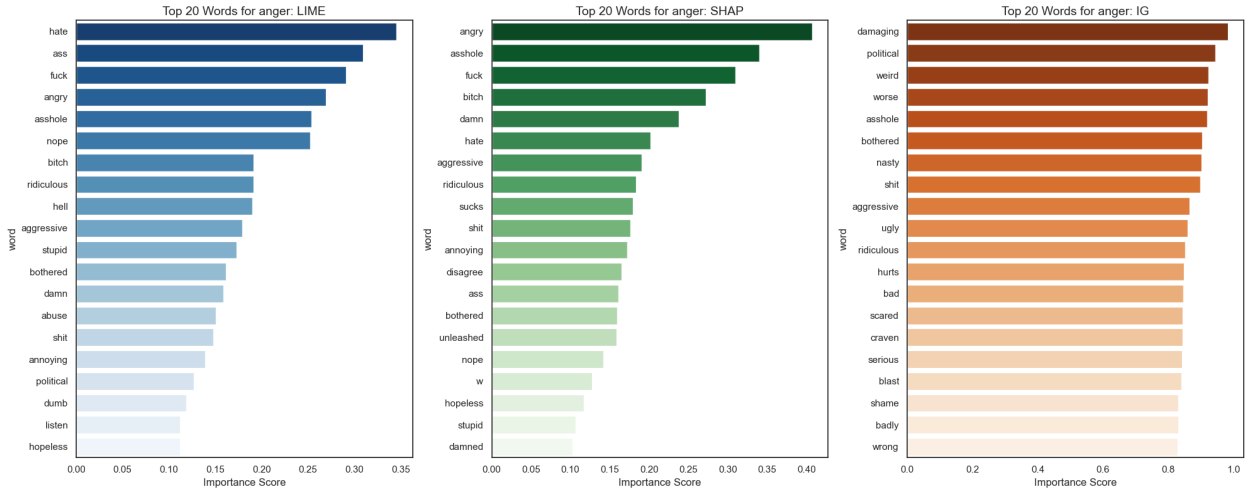

Figure 4: Top 20 tokens for the target label "anger" sorted by importance score evaluated by: *Left:* LIME. *Center:* SHAP. *Right:* Integrated Gradients.

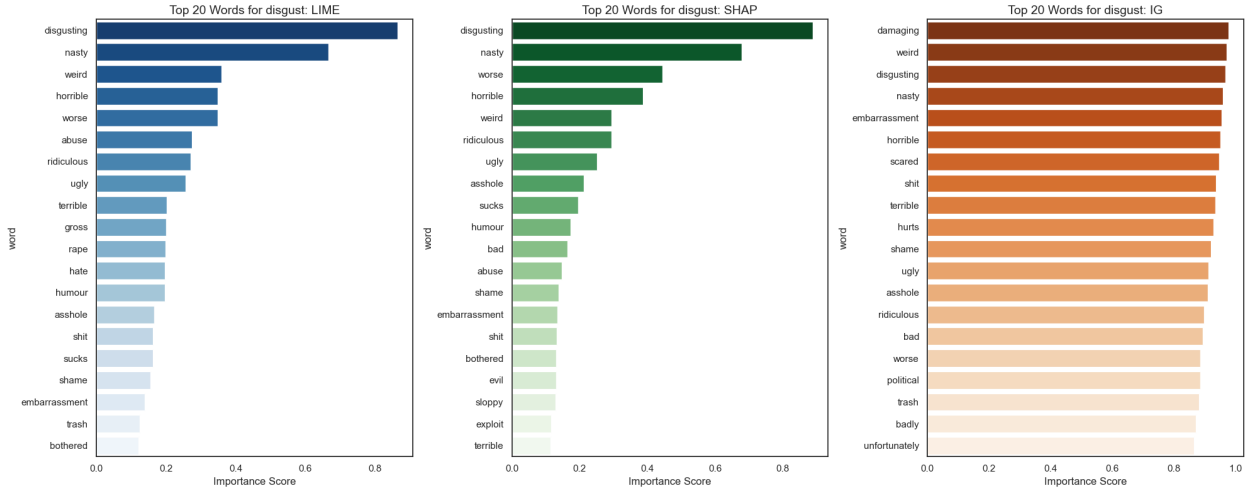

Figure 5: Top 20 tokens for the target label "disgust" sorted by importance score evaluated by: *Left:* LIME. *Center:* SHAP. *Right:* Integrated Gradients.

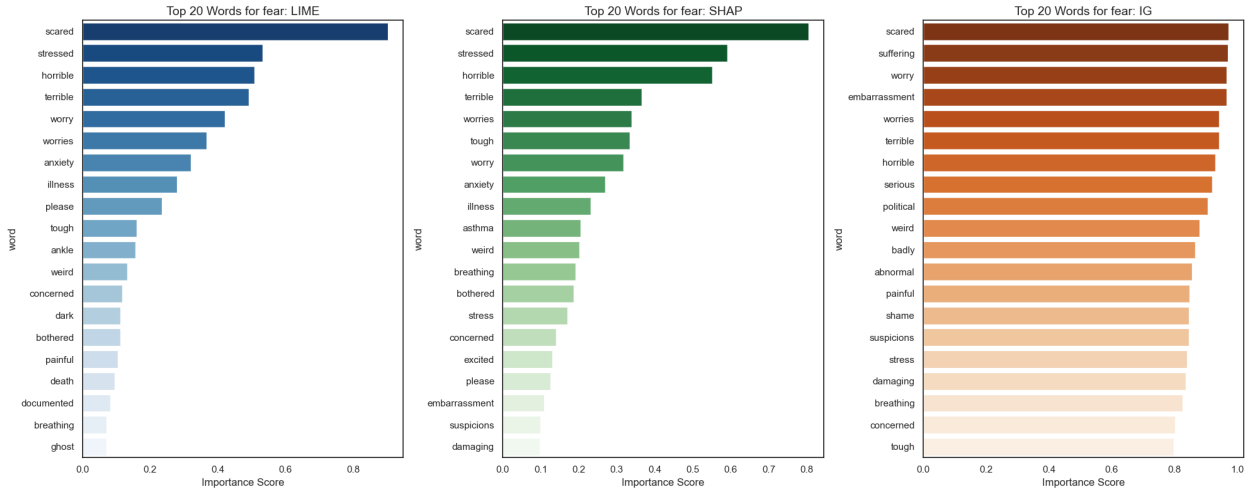

Figure 6: Top 20 tokens for the target label "fear" sorted by importance score evaluated by: *Left:* LIME. *Center:* SHAP. *Right:* Integrated Gradients.

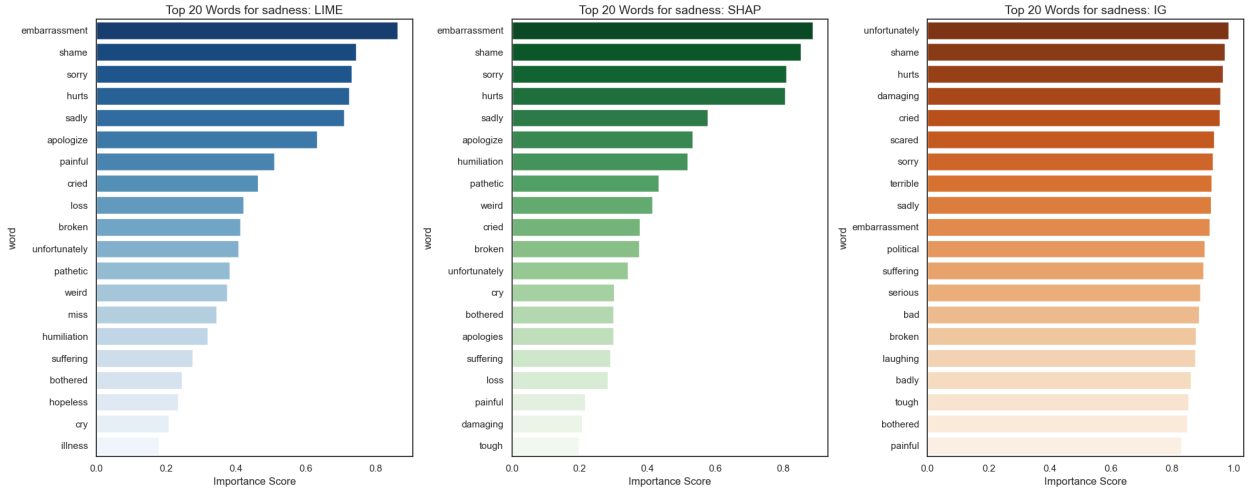

Figure 7: Top 20 tokens for the target label "sadness" sorted by importance score evaluated by: *Left:* LIME. *Center:* SHAP. *Right:* Integrated Gradients.

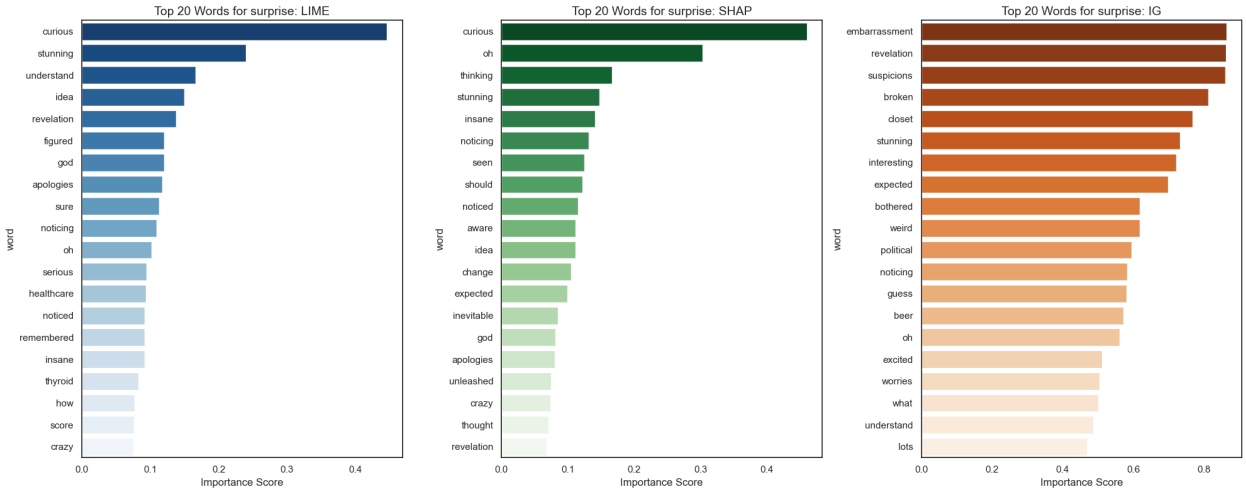

Figure 8: Top 20 tokens for the target label "surprise" sorted by importance score evaluated by: *Left:* LIME. *Center:* SHAP. *Right:* Integrated Gradients.

A.2. Jaccard Similarity Analysis

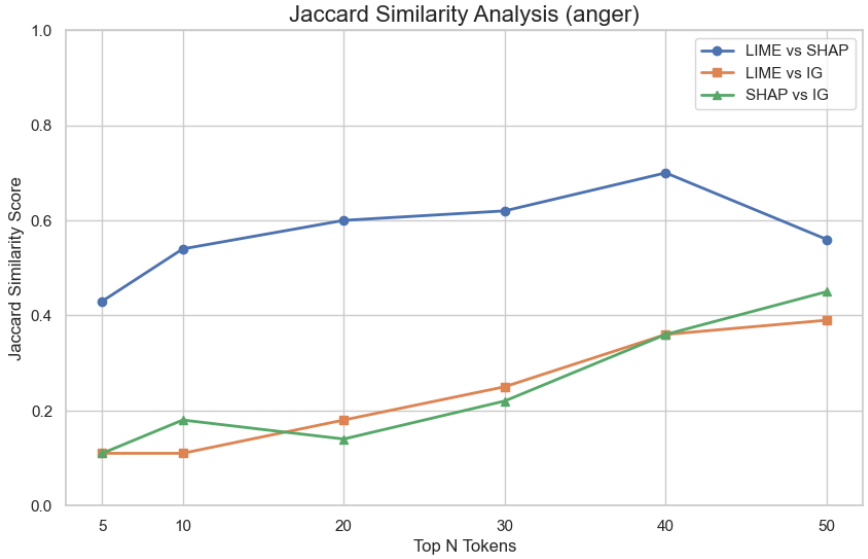

Figure 9: Jaccard Similarity Index for top-n tokens by importance score between LIME and SHAP in blue, LIME and IG in orange, and SHAP and IG in green, for the label "anger".

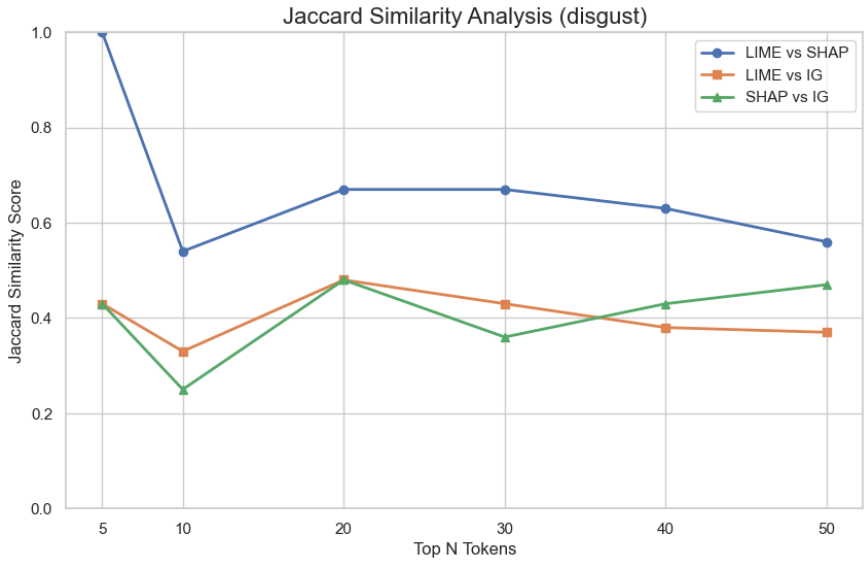

Figure 10: Jaccard Similarity Index for top-n tokens by importance score between LIME and SHAP in blue, LIME and IG in orange, and SHAP and IG in green, for the label "disgust".

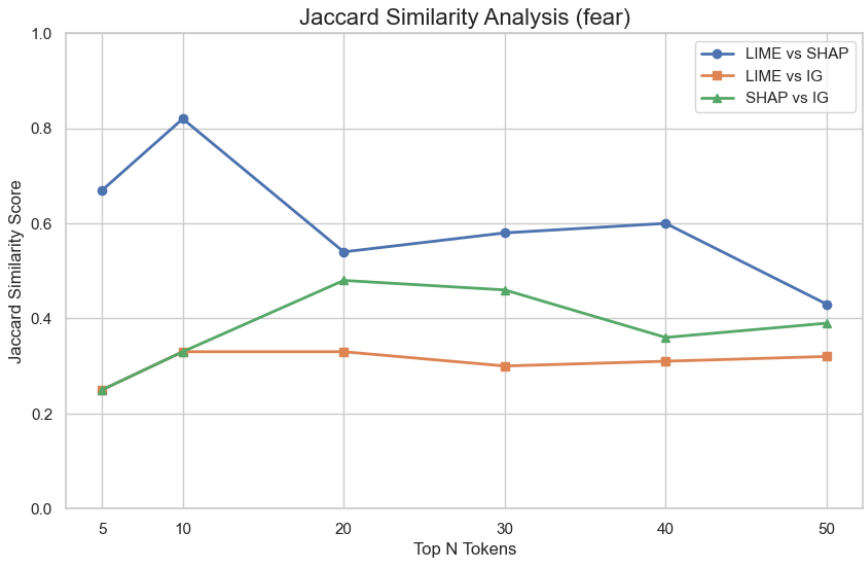

Figure 11: Jaccard Similarity Index for top-n tokens by importance score between LIME and SHAP in blue, LIME and IG in orange, and SHAP and IG in green, for the label "fear".

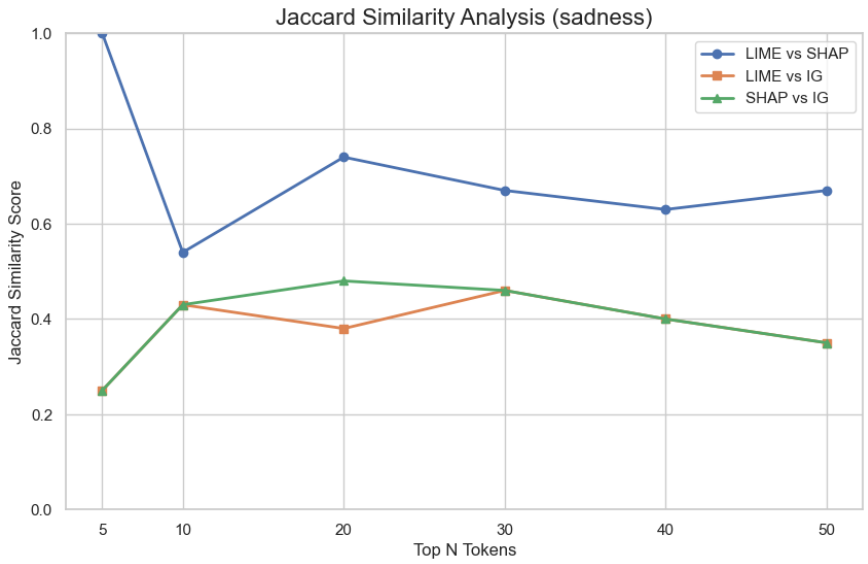

Figure 12: Jaccard Similarity Index for top-n tokens by importance score between LIME and SHAP in blue, LIME and IG in orange, and SHAP and IG in green, for the label "sadness".

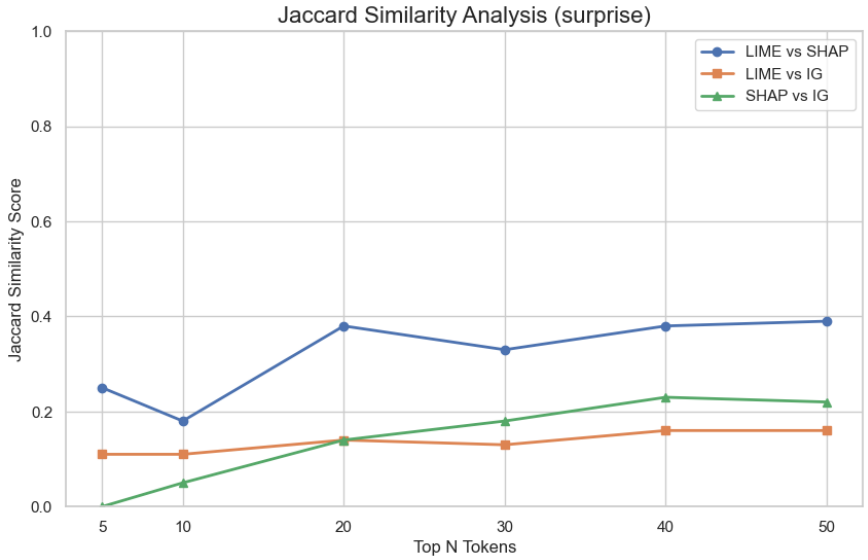

Figure 13: Jaccard Similarity Index for top-n tokens by importance score between LIME and SHAP in blue, LIME and IG in orange, and SHAP and IG in green, for the label "surprise".

A.3. 3D Importance Plots

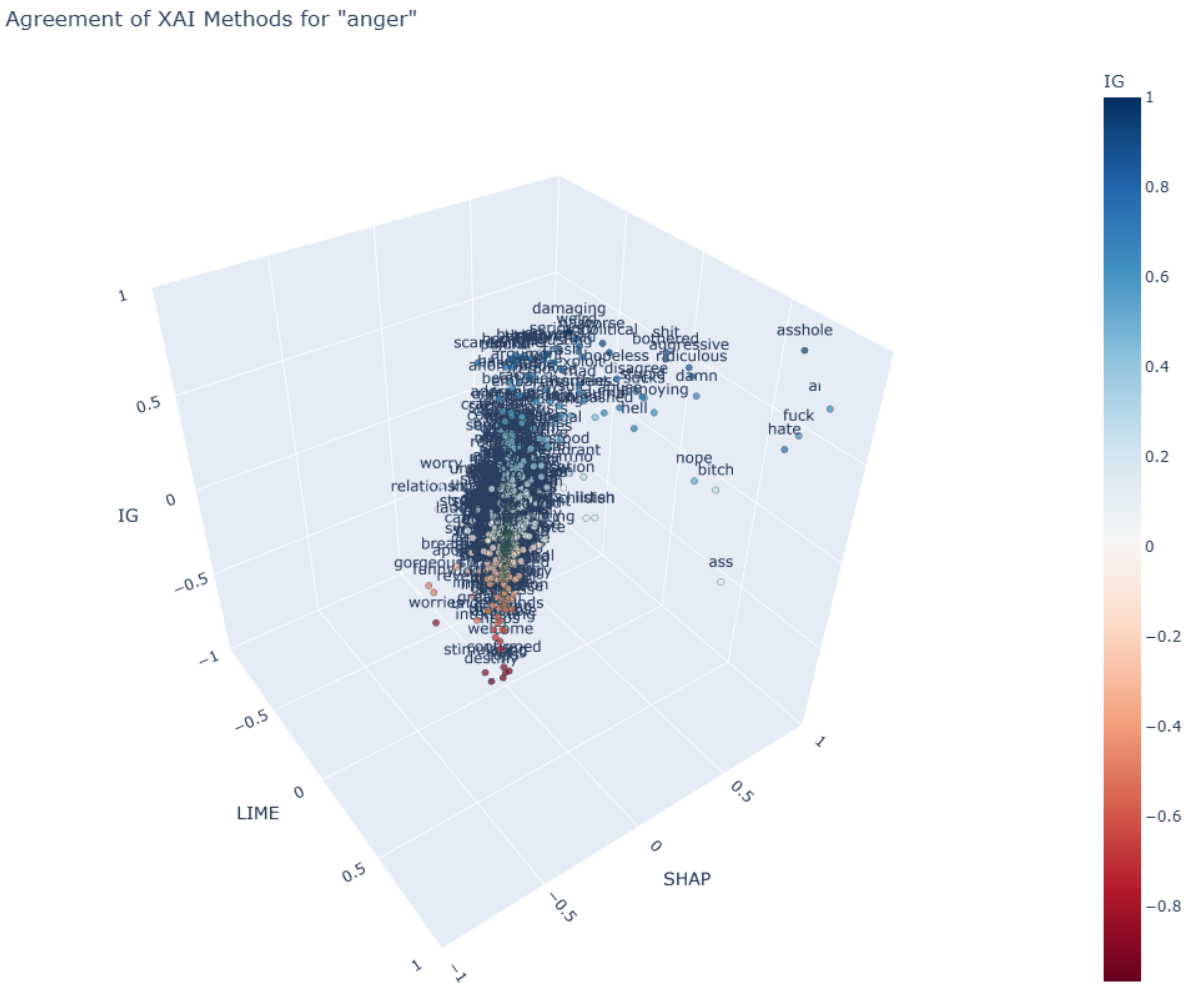

Figure 14: 3D scatter plot for the label "anger" with IG importance scores as the vertical axis, LIME importance scores as the axis along the depth of the plot, and SHAP importance scores as the axis spanning the width of the plot. The colour gradient follows the IG importance scores.

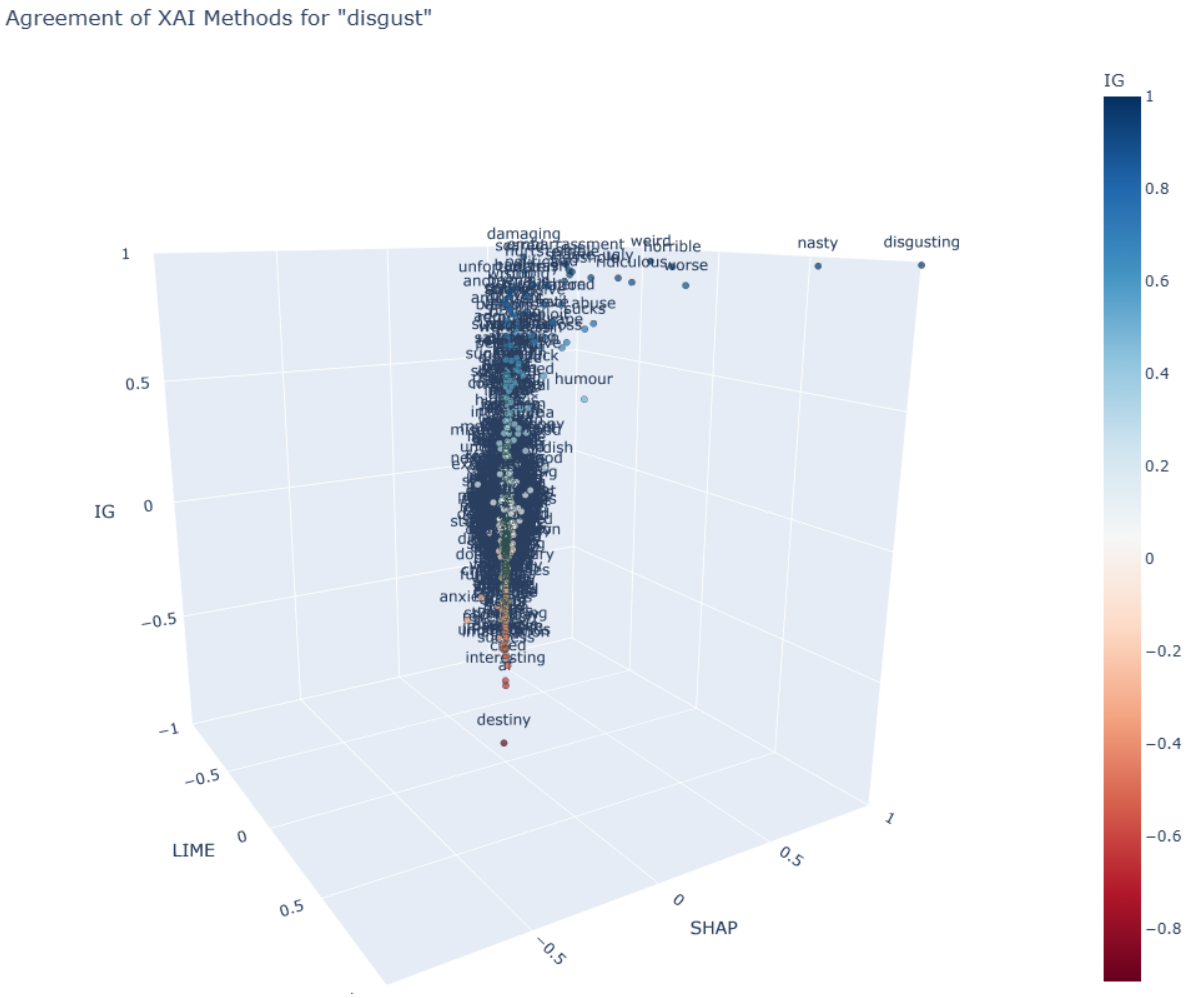

Figure 15: 3D scatter plot for the label "disgust" with IG importance scores as the vertical axis, LIME importance scores as the axis along the depth of the plot, and SHAP importance scores as the axis spanning the width of the plot. The colour gradient follows the IG importance scores.

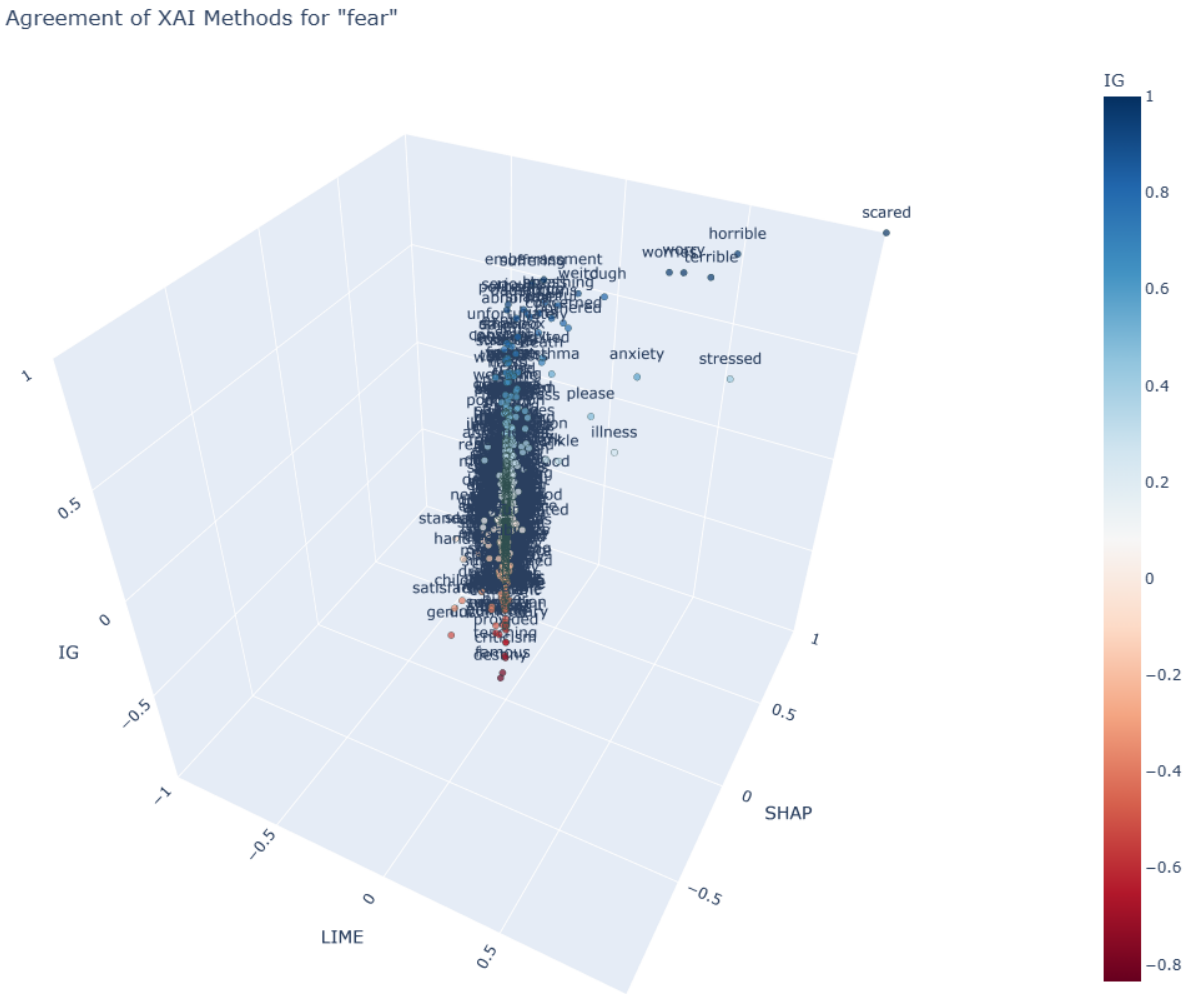

Figure 16: 3D scatter plot for the label "fear" with IG importance scores as the vertical axis, LIME importance scores as the axis along the depth of the plot, and SHAP importance scores as the axis spanning the width of the plot. The colour gradient follows the IG importance scores.

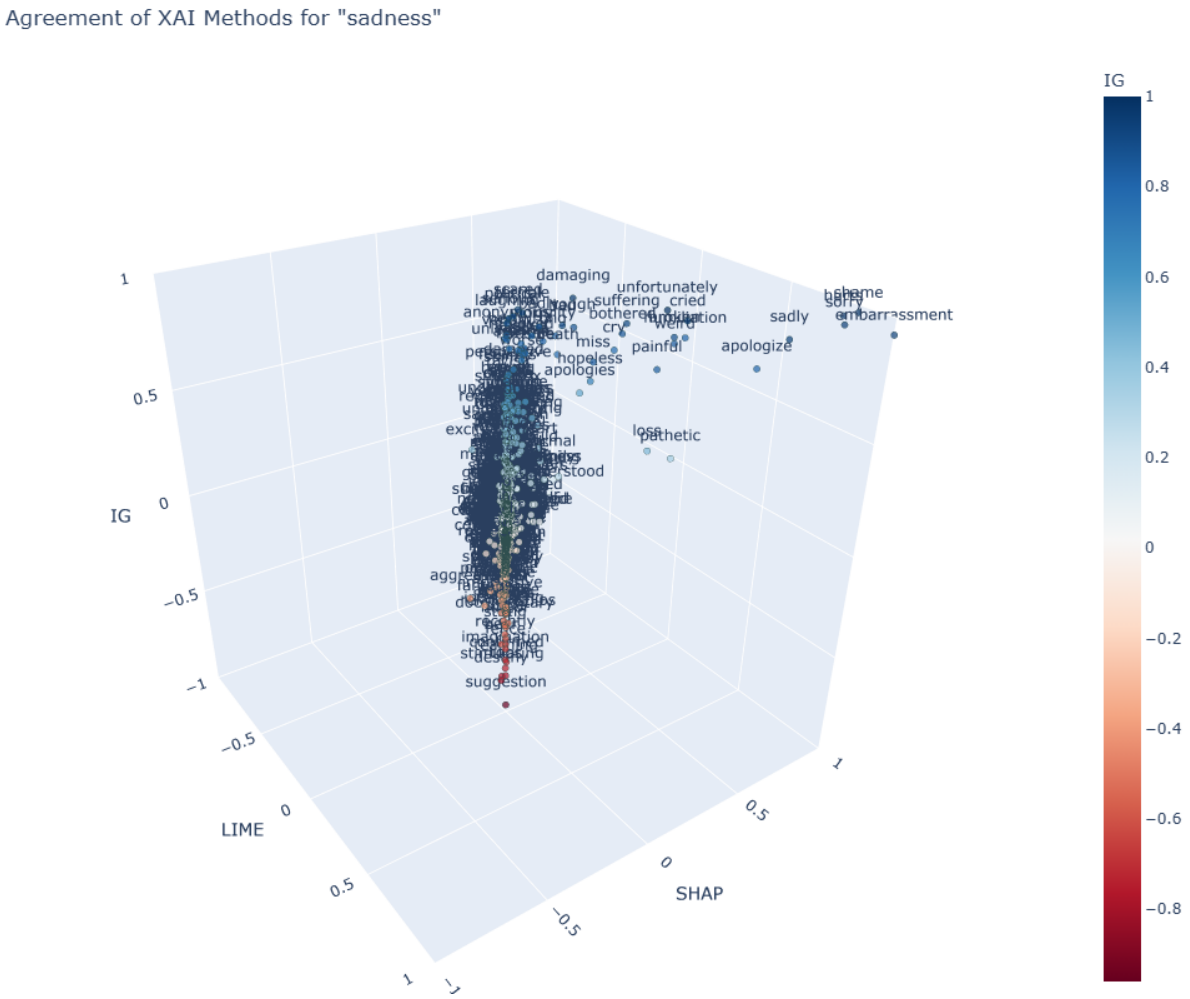

Figure 17: 3D scatter plot for the label "sadness" with IG importance scores as the vertical axis, LIME importance scores as the axis along the depth of the plot, and SHAP importance scores as the axis spanning the width of the plot. The colour gradient follows the IG importance scores.

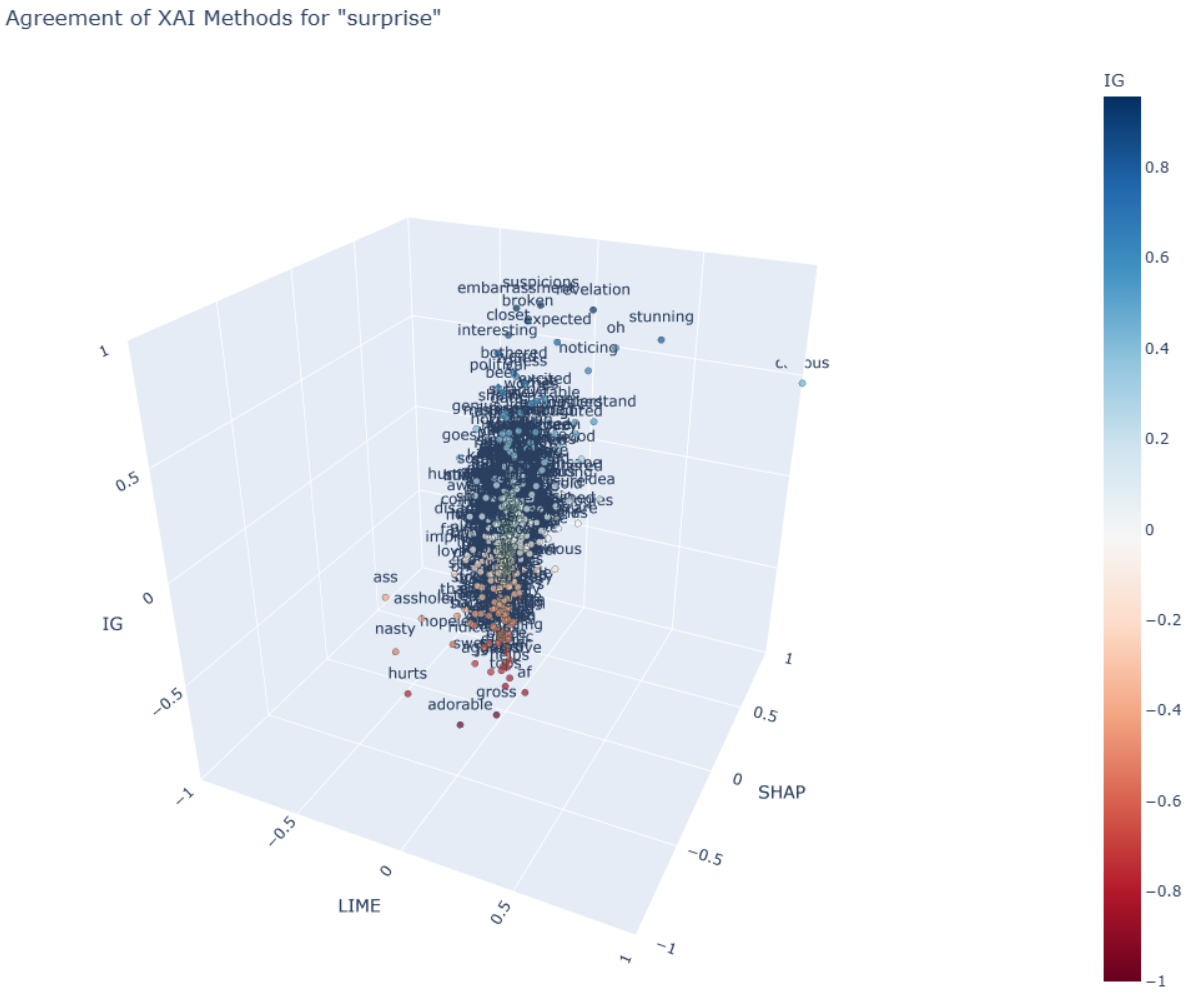

Figure 18: 3D scatter plot for the label "surprise" with IG importance scores as the vertical axis, LIME importance scores as the axis along the depth of the plot, and SHAP importance scores as the axis spanning the width of the plot. The colour gradient follows the IG importance scores.